

SBI CLERK 2026 — ALL IMPORTANT FORMULAS

Every formula you need | With examples | Quick recall format

Percentage | Profit/Loss | SI/CI | Time-Work | Speed-Distance | Ratio | Average | Algebra

1. PERCENTAGE

What to find	Formula	Example
x% of y	$y \times x \div 100$	e.g. 18% of 550 = $550 \times 18 \div 100 = 99$
A is what % of B	$(A \div B) \times 100$	e.g. What % is 45 of 180? = $45 \div 180 \times 100 = 25\%$
% increase	$(\text{New} - \text{Old}) \div \text{Old} \times 100$	e.g. Old=200, New=250 → $(250-200) \div 200 \times 100 = 25\%$
% decrease	$(\text{Old} - \text{New}) \div \text{Old} \times 100$	e.g. Old=500, New=400 → $(500-400) \div 500 \times 100 = 20\%$
Value after x% increase	$\text{Original} \times (100+x) \div 100$	e.g. 500 increased by 20% = $500 \times 120 \div 100 = 600$
Value after x% decrease	$\text{Original} \times (100-x) \div 100$	e.g. 500 decreased by 20% = $500 \times 80 \div 100 = 400$
Successive % changes a then b	$\text{Net \%} = a + b + (a \times b) \div 100$	e.g. Up 20% then down 20% = $20 + (-20) + (20 \times -20) \div 100 = -4\%$ (net loss)
Find original before x% increase	$\text{Original} = \text{Value} \times 100 \div (100+x)$	e.g. After 25% increase price is 500 → $\text{Original} = 500 \times 100 \div 125 = 400$
Price rises x%, how much less to consume to keep expenditure same	$\text{Reduction} = x \div (100+x) \times 100 \%$	e.g. Price up 25% → reduce by $25 \div 125 \times 100 = 20\%$

2. PROFIT, LOSS & DISCOUNT

What to find	Formula	Example
Profit	$SP - CP$	e.g. CP=800, SP=1000 → Profit = 200
Loss	$CP - SP$	e.g. CP=1000, SP=800 → Loss = 200
Profit %	$\text{Profit} \div CP \times 100$	e.g. Profit=200, CP=800 → $200 \div 800 \times 100 = 25\%$
Loss %	$\text{Loss} \div CP \times 100$	e.g. Loss=200, CP=1000 → $200 \div 1000 \times 100 = 20\%$
SP (given CP and Profit %)	$CP \times (100+P\%) \div 100$	e.g. CP=1200, P=25% → SP = $1200 \times 125 \div 100 = 1500$
SP (given CP and Loss %)	$CP \times (100-L\%) \div 100$	e.g. CP=1000, L=20% → SP = $1000 \times 80 \div 100 = 800$
CP (given SP and Profit %)	$SP \times 100 \div (100+P\%)$	e.g. SP=1500, P=25% → CP = $1500 \times 100 \div 125 = 1200$

What to find	Formula	Example
CP (given SP and Loss %)	$SP \times 100 \div (100 - L\%)$	e.g. $SP=800, L=20\% \rightarrow CP = 800 \times 100 \div 80 = 1000$
Discount %	$Discount \div MP \times 100$	e.g. $MP=500, Disc=50 \rightarrow 50 \div 500 \times 100 = 10\%$
SP after discount	$MP \times (100 - D\%) \div 100$	e.g. $MP=1000, D=15\% \rightarrow SP = 1000 \times 85 \div 100 = 850$
Successive discounts d1 and d2	$Net\ discount = d1 + d2 - (d1 \times d2 \div 100)$	e.g. $30\% \text{ and } 20\% = 30 + 20 - 6 = 44\% \text{ net discount}$
SP of m articles = CP of n articles $\rightarrow$ Profit %	$Profit\ \% = (n - m) \div m \times 100$	e.g. $SP \text{ of } 10 = CP \text{ of } 12 \rightarrow Profit = (12 - 10) \div 10 \times 100 = 20\%$
Mark up x%, discount y% $\rightarrow$ Profit %	$Profit\ \% = x - y - (x \times y \div 100)$	e.g. $Mark\ up\ 40\%, discount\ 25\% \rightarrow 40 - 25 - 10 = 5\% \text{ profit}$

3. SIMPLE INTEREST & COMPOUND INTEREST

What to find	Formula	Example
Simple Interest (SI)	$SI = P \times R \times T \div 100$	e.g. $P=8000, R=6\%, T=3yr \rightarrow SI = 8000 \times 6 \times 3 \div 100 = 1440$
Amount (SI)	$A = P + SI \text{ OR } P \times (1 + RT \div 100)$	e.g. $P=8000, SI=1440 \rightarrow A = 9440$
Principal (SI)	$P = SI \times 100 \div (R \times T)$	e.g. $SI=1440, R=6\%, T=3 \rightarrow P = 1440 \times 100 \div 18 = 8000$
Rate (SI)	$R = SI \times 100 \div (P \times T)$	e.g. $SI=1440, P=8000, T=3 \rightarrow R = 1440 \times 100 \div 24000 = 6\%$
Time (SI)	$T = SI \times 100 \div (P \times R)$	e.g. $SI=1440, P=8000, R=6 \rightarrow T = 1440 \times 100 \div 48000 = 3 \text{ yr}$
Doubling time (SI)	$T = 100 \div R \text{ years}$	e.g. $R=12.5\% \rightarrow \text{doubles in } 100 \div 12.5 = 8 \text{ years}$
Tripling time (SI)	$T = 200 \div R \text{ years}$	e.g. $R=12.5\% \rightarrow \text{triples in } 200 \div 12.5 = 16 \text{ years}$
Compound Interest (CI)	$CI = P \times [(1 + R \div 100)^T - 1]$	e.g. $P=10000, R=10\%, T=2 \rightarrow CI = 10000 \times [1.21 - 1] = 2100$
Amount (CI)	$A = P \times (1 + R \div 100)^T$	e.g. $P=10000, R=10\%, T=2 \rightarrow A = 10000 \times 1.21 = 12100$
CI – SI for 2 years	$= P \times (R \div 100)^2$	e.g. $P=5000, R=8\% \rightarrow CI - SI = 5000 \times 0.0064 = 32$
CI – SI for 3 years	$= P \times R^2 \times (300 + R) \div 100^3$	e.g. Rarely asked — use CI–SI formula if needed
Half-yearly compounding	$A = P \times (1 + R \div 200)^{(2T)}$	e.g. $R=10\%, T=1yr \text{ half-yearly} \rightarrow A = P \times (1.05)^2 = 1.1025P$

4. TIME & WORK

What to find	Formula	Example
A's 1-day work	$1 \div n$ (if A finishes in n days)	e.g. A finishes in 12 days $\rightarrow$ 1-day work = $1 \div 12$
A and B together	$ab \div (a+b)$ days	e.g. A=12 days, B=18 days $\rightarrow$ Together= $12 \times 18 \div 30 = 7.2$ days
A, B, C together	$1 \div (1/a + 1/b + 1/c)$ days	e.g. A=12, B=15, C=20 $\rightarrow 1 \div (5+4+3)/60 = 60 \div 12 = 5$ days
Pipe A fills, Pipe B drains ($b > a$)	Net fill time = $ab \div (b-a)$ hours	e.g. A fills in 6h, B drains in 10h $\rightarrow$ $6 \times 10 \div (10-6) = 15$ hrs
M men do work in D days $\rightarrow$ M1 men take how many days	$M \times D = M1 \times D1$	e.g. 10 men in 12 days = 15 men in ? $\rightarrow$ $15 \times D1 = 120 \rightarrow D1 = 8$ days
Work W in D days with M men (add workers mid-way)	Total work (units) = $M \times D$ Split by phases	e.g. See each phase separately: 10 men $\times$ 5 days + 15 men $\times$ x days = total
A is twice as efficient as B	If B takes n days, A takes $n \div 2$ days	e.g. B=20 days $\rightarrow$ A=10 days $\rightarrow$ Together= $10 \times 20 \div 30 = 6.67$ days
A can do in x days, leaves after y days, B finishes remaining	Work done by A = $y \div x$ Remaining = $1 - y \div x$ B's time = Remaining $\times b$	e.g. A=12 days leaves after 4 days $\rightarrow$ Done= $4 \div 12 = 1 \div 3$ Remaining= $2 \div 3 \rightarrow B=18$ days $\rightarrow$ B needs $18 \times 2 \div 3 = 12$ days

5. SPEED, DISTANCE & TIME

What to find	Formula	Example
Basic relation	Speed = Distance $\div$ Time	e.g. D=120km, T=2hr $\rightarrow$ S=60 km/h
km/h $\rightarrow$ m/s	Multiply by $5 \div 18$	e.g. 72 km/h = $72 \times 5 \div 18 = 20$ m/s
m/s $\rightarrow$ km/h	Multiply by $18 \div 5$	e.g. 20 m/s = $20 \times 18 \div 5 = 72$ km/h
Average speed (equal distances)	$2ab \div (a+b)$	e.g. Going 60, return 40 $\rightarrow$ Avg = $2 \times 60 \times 40 \div 100 = 48$ km/h
Relative speed — OPPOSITE direction	$v1 + v2$	e.g. Train A=60, Train B=40 $\rightarrow$ Rel speed=100 km/h
Relative speed — SAME direction	$ v1 - v2 $	e.g. Train A=60, Train B=40 $\rightarrow$ Rel speed=20 km/h
Train crosses a POLE	Speed = Train length $\div$ Time	e.g. Train 200m, time 10sec $\rightarrow$ Speed= $200 \text{m} \div 10 \text{sec} = 20 \text{m/s} = 72 \text{km/h}$
Train crosses a PLATFORM	Speed = (Train + Platform) $\div$ Time	e.g. Train 300m, Platform 200m, time 25s $\rightarrow S = 500 \div 25 = 20 \text{m/s} = 72 \text{km/h}$
Two trains cross each other (opposite direction)	Time = $(L1 + L2) \div (v1 + v2)$	e.g. 300m+200m, speeds 60+40 km/h $\rightarrow 500 \text{m} \div (100 \times 5 \div 18) = 18 \text{sec}$
Boat UPSTREAM speed	Boat speed – Stream speed	e.g. Boat=20, Stream=5 $\rightarrow$ Upstream=15 km/h
Boat DOWNSTREAM speed	Boat speed + Stream speed	e.g. Boat=20, Stream=5 $\rightarrow$ Downstream=25 km/h

What to find	Formula	Example
Stream speed	$(\text{Downstream} - \text{Upstream}) \div 2$	e.g. Down=25, Up=15 $\rightarrow$ Stream=(25-15) $\div$ 2=5 km/h
Boat speed in still water	$(\text{Downstream} + \text{Upstream}) \div 2$	e.g. Down=25, Up=15 $\rightarrow$ Boat=(25+15) $\div$ 2=20 km/h

6. AVERAGE

What to find	Formula	Example
Average	Sum of all values $\div$ Count of values	e.g. Sum=400, Count=8 $\rightarrow$ Average=50
Sum from average	Sum = Average $\times$ Count	e.g. Average=40, Count=5 $\rightarrow$ Sum=200
Number removed — new average given	Removed = Old sum - New sum = (Old avg $\times$ n) - (New avg $\times$ (n-1))	e.g. Avg of 5=40 $\rightarrow$ Sum=200. Remove one, avg=36 $\rightarrow$ New sum=144. Removed=200-144=56
Number added — new average given	Added = New sum - Old sum = (New avg $\times$ (n+1)) - (Old avg $\times$ n)	e.g. Avg of 5=40 $\rightarrow$ Sum=200. Add one, avg=42 $\rightarrow$ New sum=252. Added=252-200=52
Average of 1 to n natural numbers	$(n+1) \div 2$	e.g. Average of 1 to 100 = 101 $\div$ 2 = 50.5
Average of first n odd numbers	n	e.g. Average of first 7 odd numbers = 7
Average of first n even numbers	n+1	e.g. Average of first 7 even numbers = 8
Weighted average	$(w1 \times x1 + w2 \times x2 + \dots) \div (w1 + w2 + \dots)$	e.g. 30 students avg 50, 20 students avg 60 $\rightarrow$ Wt avg=(1500+1200) $\div$ 50=54

7. RATIO, PROPORTION & PARTNERSHIP

What to find	Formula	Example
If a:b and b:c $\rightarrow$ a:b:c	Make b common. Multiply to equate.	e.g. a:b=3:4, b:c=2:3 $\rightarrow$ b=4 and b=2 $\rightarrow$ $\times 2$: b:c=4:6 $\rightarrow$ a:b:c=3:4:6
Two numbers in ratio a:b, sum=S	First = $aS \div (a+b)$, Second = $bS \div (a+b)$	e.g. Ratio 3:5, sum=80 $\rightarrow$ First=3 $\times$ 80 $\div$ 8=30, Second=50
HCF=h, ratio a:b $\rightarrow$ LCM	Numbers = ah and bh. LCM = $ah \times bh \div h = abh$	e.g. HCF=12, ratio 3:5 $\rightarrow$ Numbers=36,60 $\rightarrow$ LCM=180
Partnership — same time	Profit ratio = Investment1 : Investment2	e.g. A=30000, B=50000 $\rightarrow$ ratio=3:5 $\rightarrow$ A gets 3 $\div$ 8 of profit
Partnership — different time	Profit ratio = P1 $\times$ T1 : P2 $\times$ T2	e.g. A=20000 for 6mo, B=30000 for 4mo $\rightarrow$ 120000:120000 = 1:1
Alligation / Mixture rule	Qty1 : Qty2 = (d-m) : (m-c) d=dearer price, c=cheaper price, m=mean price	e.g. Milk at Rs.20, Water at Rs.0, mix at Rs.15 $\rightarrow$ Water:Milk = (20-15):(15-0) = 5:15 = 1:3

What to find	Formula	Example
Mixture — remove v litres from V litres, n times	Final conc. = $C \times (1 - v \div V)^n$	e.g. 100L pure milk. Remove 10L, add water 3 times → Milk left = $100 \times (0.9)^3 = 72.9L$

8. QUADRATIC EQUATIONS (SBI Clerk Mains — 5 Q)

Step	Method	Example
Step 1 — Identify coefficients	$ax^2 + bx + c = 0$	e.g. $x^2 - 5x + 6 = 0 \rightarrow a=1, b=-5, c=6$
Step 2 — Find two numbers	Product = c, Sum = -b (for a=1; when a≠1, product=c÷a, sum=-b÷a)	e.g. Product=6, Sum=5 → numbers are 2 and 3 ($2 \times 3=6, 2+3=5$)
Step 3 — Write roots	$(x - r1)(x - r2) = 0 \rightarrow x = r1 \text{ or } r2$	e.g. $(x-2)(x-3)=0 \rightarrow x=2 \text{ or } x=3$
Step 4 — Compare x and y	Check ALL pairs (xi, yj) and determine: $x > y$ if ALL $x_i > y_j$ $x \leq y$ if in all pairs $x_i \leq y_j$	e.g. $x=2,3$ and $y=3,4$ $(2,3)x < y$ $(2,4)x < y$ $(3,3)x=y$ $(3,4)x < y \rightarrow$ All cases $x \leq y$ ✓
Discriminant (when factoring is hard)	Roots = $(-b \pm \sqrt{(b^2-4ac)}) \div 2a$	e.g. $2x^2-7x+3=0 \rightarrow \text{disc}=49-24=25$ $x=(7 \pm 5) \div 4 \rightarrow x=3 \text{ or } x=0.5$

$x > y$: ALL pairs $x_i > y_j$	$x < y$: ALL pairs $x_i < y_j$	$x \geq y$: All $x_i \geq y_j$, at least one equal	$x \leq y$: All $x_i \leq y_j$, at least one equal	Cannot determine: Mixed results
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9. NUMBER SERIES — COMMON PATTERNS

Pattern type	How to identify	Example
Multiply by fixed number	Each term = previous $\times n$	2, 6, 18, 54, 162 → $\times 3$ each time
Add fixed number	Difference between terms is constant	5, 9, 13, 17, 21 → $+4$ each time
Difference increasing by fixed amount	D: 2,4,6,8,10 or 3,5,7,9	3, 7, 13, 21, 31 → diff 4,6,8,10 → next diff 12 → next term 43
Perfect squares	1,4,9,16,25,36,49,64... → n^2	1, 4, 9, 16, 25, 36, 49 → ? = 64 (8^2)
Perfect cubes	1,8,27,64,125,216 → n^3	1, 8, 27, 64, 125, ? = 216 (6^3)
Fibonacci	Each term = sum of previous two	1, 1, 2, 3, 5, 8, 13, 21, 34...
$\times 2$ then $+1$ alternating	Mixed operations	3, 7, 15, 31, 63 → $\times 2+1$ each time
Wrong number	Find which term breaks the pattern	3, 9, 27, 81, 243, 729, 2196 → should be 2187 ($\times 3$). Wrong: 2196

10. AGES

What to find	Formula / Method	Example
Father is n times son's age, after t years ratio a:b	Let son=x. Father=nx. $(nx+t) \div (x+t) = a \div b \rightarrow$ solve for x	e.g. Father=4x, son=x. After 5yr ratio 3:1 $(4x+5) \div (x+5) = 3 \rightarrow 4x+5=3x+15 \rightarrow x=10$
A is p years older than B. Ratio of ages now a:b	Let B=x. A=x+p. $(x+p) \div x = a \div b \rightarrow$ solve	e.g. A is 10 older than B. Ratio now 2:1 $(x+10) \div x = 2 \rightarrow x+10=2x \rightarrow x=10. A=20.$
n years ago ratio was a:b, now ratio is c:d	Let current ages = cx and dx. $cx - n \div dx - n = a \div b \rightarrow$ solve	e.g. Now 3:2, 5 yrs ago 2:1 $3x-5 \div 2x-5 = 2 \div 1 \rightarrow 3x-5=4x-10 \rightarrow x=5$ Ages: 15 and 10

11. SIMPLIFICATION SPEED TRICKS

Trick	Method	Example
Multiply any 2-digit number by 11	Keep first add both digits keep last If middle sum ≥ 10 , carry over	54×11 : $5 (5+4) 4 = 594$ 67×11 : $6 (6+7) 7 = 6 13 7 = 737$
Square a number ending in 5	$n5^2 = n \times (n+1)$ followed by 25	$35^2 = 3 \times 4 = 12$, append 25 $\rightarrow 1225$ $75^2 = 7 \times 8 = 56$, append 25 $\rightarrow 5625$
Divide by 5	Multiply by 2, then divide by 10	$840 \div 5 = 1680 \div 10 = 168$
Divide by 25	Multiply by 4, then divide by 100	$1750 \div 25 = 7000 \div 100 = 70$
Divide by 125	Multiply by 8, then divide by 1000	$2500 \div 125 = 20000 \div 1000 = 20$
Find 15% quickly	10% of number + half of that	15% of 840 = $84 + 42 = 126$
Find 12.5%	Divide by 8 (since $12.5\% = 1/8$)	12.5% of 560 = $560 \div 8 = 70$
Approximate large multiplication	Round to nearest clean number, then adjust	$399.8 \times 20.1 \div 4.98 \approx 400 \times 20 \div 5 = 1600$
BODMAS order — never get this wrong	Brackets $\rightarrow$ Of $\rightarrow$ Division $\rightarrow$ Multiplication $\rightarrow$ Addition $\rightarrow$ Subtraction D and M are left-to-right. A and S are left-to-right.	$4800 \div 24 \times 5 - 400 = 200 \times 5 - 400 = 1000 - 400 = 600$ ($\div$ before $\times$ is wrong — do left to right)

12. MUST-MEMORISE VALUES

Squares (1–25):

$1^2=1$	$2^2=4$	$3^2=9$	$4^2=16$	$5^2=25$
$6^2=36$	$7^2=49$	$8^2=64$	$9^2=81$	$10^2=100$
$11^2=121$	$12^2=144$	$13^2=169$	$14^2=196$	$15^2=225$
$16^2=256$	$17^2=289$	$18^2=324$	$19^2=361$	$20^2=400$
$21^2=441$	$22^2=484$	$23^2=529$	$24^2=576$	$25^2=625$

Cubes (1–15):

$1^3=1$	$2^3=8$	$3^3=27$	$4^3=64$	$5^3=125$
$6^3=216$	$7^3=343$	$8^3=512$	$9^3=729$	$10^3=1000$
$11^3=1331$	$12^3=1728$	$13^3=2197$	$14^3=2744$	$15^3=3375$

Fractions as Percentages (use these to avoid division):

1/2=50%	1/3=33.33%	1/4=25%	1/5=20%	1/6=16.67%
1/7=14.28%	1/8=12.5%	1/9=11.11%	1/10=10%	1/11=9.09%
1/12=8.33%	2/3=66.67%	3/4=75%	2/5=40%	3/5=60%
4/5=80%	3/8=37.5%	5/8=62.5%	7/8=87.5%	5/6=83.33%

Square Roots ($\sqrt{1}$ to $\sqrt{400}$):

$\sqrt{1}=1$	$\sqrt{4}=2$	$\sqrt{9}=3$	$\sqrt{16}=4$	$\sqrt{25}=5$
$\sqrt{36}=6$	$\sqrt{49}=7$	$\sqrt{64}=8$	$\sqrt{81}=9$	$\sqrt{100}=10$
$\sqrt{121}=11$	$\sqrt{144}=12$	$\sqrt{169}=13$	$\sqrt{196}=14$	$\sqrt{225}=15$
$\sqrt{256}=16$	$\sqrt{289}=17$	$\sqrt{324}=18$	$\sqrt{361}=19$	$\sqrt{400}=20$

These formulas cover 100% of the Quantitative section in SBI Clerk.
You do NOT need to memorise all of them tonight — you already know most.
Just glance through once. Sleep by 10 PM. All the best.